

DevOps Lab L1 Assignment Submission

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GitHub Repository: <https://github.com/aryan20s/Devops-Lab-L1-Submission>

1. Git Workflow & Collaboration (Task 1)

Task 1.1: Initialize Git Repository and Push Main

Initialized a new Git repository for the "Hello World" Python Flask application, added the initial code, committed it, and pushed to the `main` branch.

```
PS C:\Users\User\Programming\Random\Tools\23070122055_aryansrivastava\23070122055_aryansrivastava> git remote add origin https://github.com/aryan20s/Devops-Lab-L1-Submission.git
PS C:\Users\User\Programming\Random\Tools\23070122055_aryansrivastava\23070122055_aryansrivastava> git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (9/9), 827 bytes | 827.00 KiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/aryan20s/Devops-Lab-L1-Submission.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Task 1.2: Feature Branch Creation & Modification

Created a new branch named `feature/user-auth`. Made modifications to the Flask app to add authentication logic, committed the changes, and pushed the branch to the remote repository.

```
PS C:\Users\User\Programming\Random\Tools\23070122055_aryansrivastava\23070122055_aryansrivastava> git checkout feature/user-auth
Switched to branch 'feature/user-auth'
PS C:\Users\User\Programming\Random\Tools\23070122055_aryansrivastava\23070122055_aryansrivastava> git push -u origin feature/user-auth
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 416 bytes | 416.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/user-auth' on GitHub by visiting:
remote:   https://github.com/aryan20s/Devops-Lab-L1-Submission/pull/new/feature/user-auth
remote:
To https://github.com/aryan20s/Devops-Lab-L1-Submission.git
 * [new branch]      feature/user-auth -> feature/user-auth
branch 'feature/user-auth' set up to track 'origin/feature/user-auth'.
```

Task 1.3: Simulate and Resolve Merge Conflict

Simulated a merge conflict by modifying the same line on the `main` branch and attempting to merge `feature/user-auth` into `main`.

Merge Conflict Detected in Code:

```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route('/')
6 def hello_world():
7     <<<<<<< HEAD
8     print("Hello world accessed!")
9     =====
10    print("User authenticated successfully!")
11 >>>>>>> feature/user-auth
12    return 'Hello World!'
13
14 if __name__ == '__main__':
15     app.run(debug=True, host='0.0.0.0', port=5000)
16
```

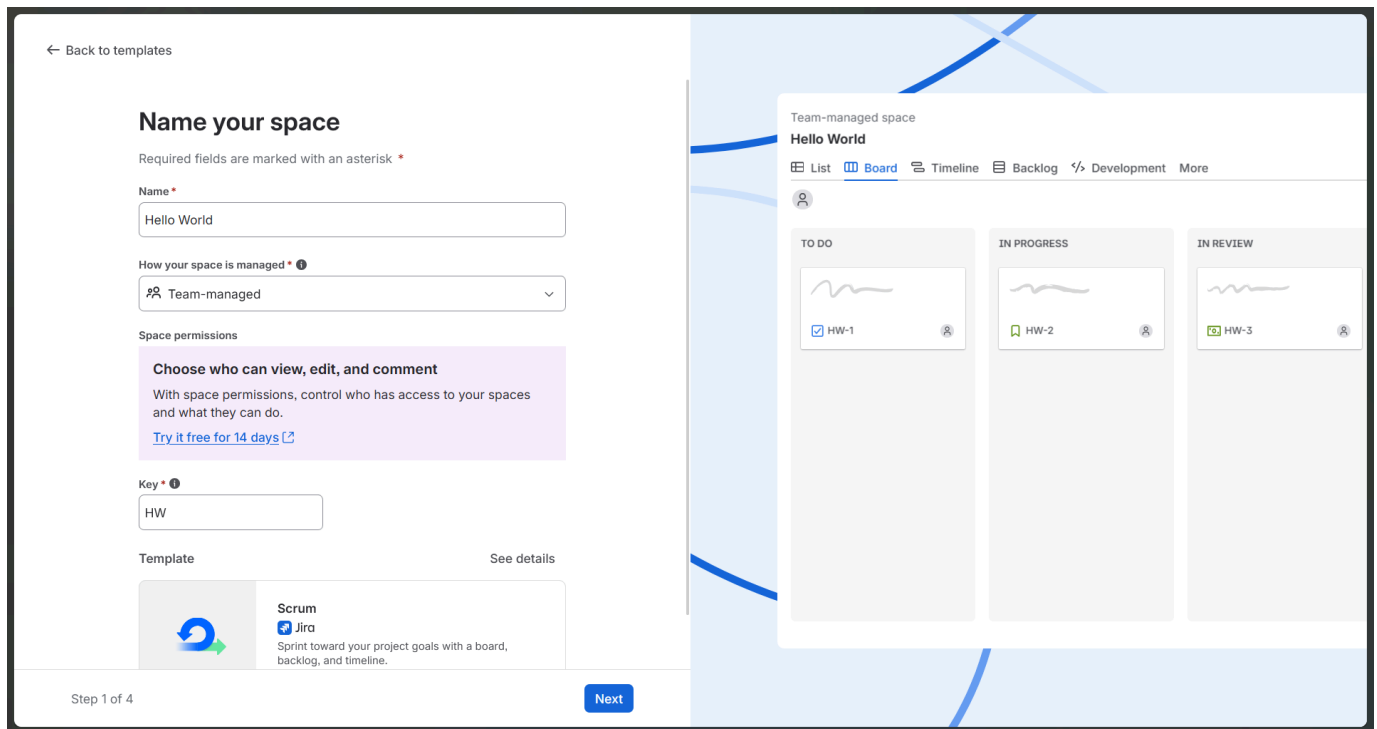
Manual Conflict Resolution and Push: Manually resolved the conflict, committed the resolution, and successfully pushed the updated `main` branch.

```
PS C:\Users\User\Programming\Random\Tools\23070122055_aryansrivastava\23070122055_aryansrivastava> git add app/app.py
PS C:\Users\User\Programming\Random\Tools\23070122055_aryansrivastava\23070122055_aryansrivastava> git commit -m "Merge branch 'feature/user-auth' into main: Resolved conflict"
[main 9c64aac] Merge branch 'feature/user-auth' into main: Resolved conflict
PS C:\Users\User\Programming\Random\Tools\23070122055_aryansrivastava\23070122055_aryansrivastava> git push origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 451 bytes | 451.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/aryan20s/Devops-Lab-L1-Submission.git
 868eaf0..9c64aac main -> main
```

2. Jira Project & Issue Tracking (Task 2)

Task 2.1: Scrum Project Creation

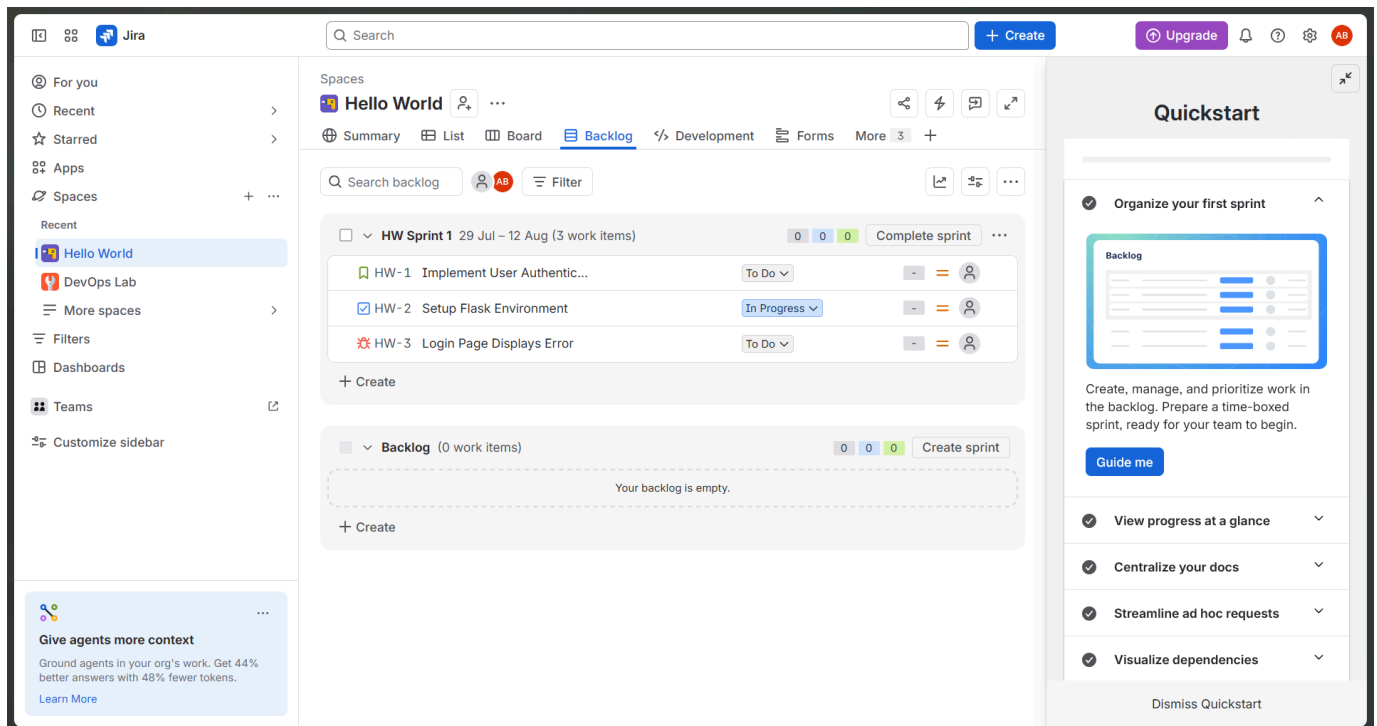
Created a new Scrum project in Jira Cloud named "Hello World".



Task 2.2: Issue Creation

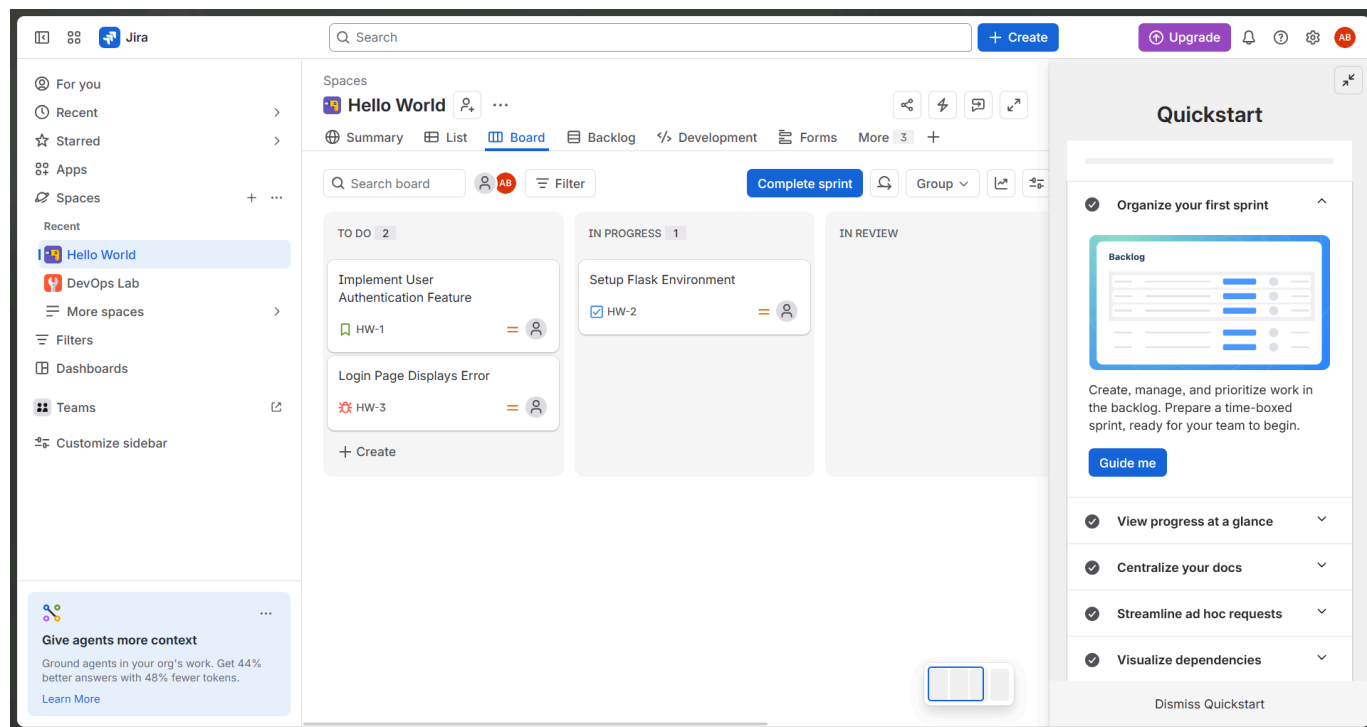
Created three distinct issues as requested:

- **Story:** Implement User Authentication Feature
- **Task:** Setup Flask Environment
- **Bug:** Login Page Displays Error



Task 2.3: Sprint Board Management

Moved the "Setup Flask Environment" task from "To Do" to the "In Progress" column on the project's board.



3. Basic Containerization (Docker) & Jenkins (Task 3)

Task 3.1: Dockerfile & Container Verification

Created a `Dockerfile` for the Python Flask application to run on port 5000. Built the Docker image locally and started the container.

Docker Build and Run:

```
user@DESKTOP-OT34PJ8:/mnt/c/Users/User/Programming/Random/Tools/23070122055_aryansrivastava/23070122055_aryansrivastava/app$ sudo docker build -t flask-hello-world .
[sudo] password for user:
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.096kB
Step 1/7 : FROM python:3.9-slim
3.9-slim: Pulling from library/python
38513bd72563: Pulling fs layer
b3ec39b36ae8: Pulling fs layer
fc7443084902: Pulling fs layer
ea56f685404a: Pulling fs layer
ea56f685404a: Waiting
b3ec39b36ae8: Verifying Checksum
b3ec39b36ae8: Download complete
ea56f685404a: Verifying Checksum
ea56f685404a: Download complete
fc7443084902: Verifying Checksum
fc7443084902: Download complete
38513bd72563: Verifying Checksum
38513bd72563: Download complete
38513bd72563: Pull complete
b3ec39b36ae8: Pull complete
fc7443084902: Pull complete
ea56f685404a: Pull complete
Digest: sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acdale13cf1731b1b
Status: Downloaded newer image for python:3.9-slim
--> 085da638e1b8
Step 2/7 : WORKDIR /app
--> Running in 1e46f39ca752
--> Removed intermediate container 1e46f39ca752
--> 1f1biae727cf
Step 3/7 : COPY requirements.txt .
--> f7346c215722
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
--> Running in fafbc393a784
Collecting Flask==3.0.0
  Downloading flask-3.0.0-py3-none-any.whl (99 kB)
    _____ 99.7/99.7 kB 7.2 MB/s eta 0:00:00
Collecting Jinja2>=3.1.2
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
Collecting blinker>=1.6.2
  Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Collecting itsdangerous>=2.1.2
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting importlib-metadata>=3.6.0
  Downloading importlib_metadata-8.7.1-py3-none-any.whl (27 kB)
Collecting Werkzeug>=3.0.0
  Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
    _____ 226.5/226.5 kB 20.5 MB/s eta 0:00:00
Collecting click>=8.1.3
  Downloading click-8.1.8-py3-none-any.whl (98 kB)
    _____ 98.2/98.2 kB 23.6 MB/s eta 0:00:00
Collecting zipp>=3.20
  Downloading zipp-3.23.1-py3-none-any.whl (10 kB)
Collecting MarkupSafe>=2.0
  Downloading markupsafe-3.0.3-cp39-cp39-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (20 kB)
Successfully installed Flask-3.0.0 Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.8 blinker-1.9.0 click-8.1.8 importlib-metadata-8.7.1 itsdangerous-2.2.0 zipp-3.23.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: http
s://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
--> Removed intermediate container fafbc393a784
--> d54707904080
Step 5/7 : COPY . .
--> 55f2b67d66e6
Step 6/7 : EXPOSE 5000
--> Running in aa75a0fe98db
--> Removed intermediate container aa75a0fe98db
--> 28c9bd23a3db
Step 7/7 : CMD ["python", "app.py"]
--> Running in e7112e4fd57a
--> Removed intermediate container e7112e4fd57a
--> 443abf89fe41
Successfully built 443abf89fe41
Successfully tagged flask-hello-world:latest
user@DESKTOP-OT34PJ8:/mnt/c/Users/User/Programming/Random/Tools/23070122055_aryansrivastava/23070122055_aryansrivastava/app$ 
user@DESKTOP-OT34PJ8:/mnt/c/Users/User/Programming/Random/Tools/23070122055_aryansrivastava/23070122055_aryansrivastava/app$ sudo docker run -d -p 5000:5000 --name my-flask-app flask-hello-wor
ld
2501cc98e1d1171b08d276854af722f7733da02a43cc7e8282a8c40360707cf6
```

Verification (cURL localhost:5000): The application successfully returns "Hello World!" confirming the container is running correctly.

```
user@DESKTOP-OT34PJ8:/mnt/c/Users/User/Programming/Random/Tools/23070122055_aryansrivastava/23070122055_aryansrivastava/app$ curl http://localhost:5000; echo
Hello World!
```

Task 3.2: Jenkins Freestyle Project Setup

Set up Jenkins using a Docker container and configured a Freestyle project to pull the Git repository and execute a shell script build step.

Jenkins Docker Setup and Run Output:

```
user@DESKTOP-OT34PJB:/mnt/c/Users/User/Programming/Random/Tools/23070122055_aryansrivastava/23070122055_aryansrivastava/app$ sudo docker run -d -p 8080:8080 -p 50000:50000 --name jenkins jenkins/jenkins:ls
Unable to find image 'jenkins/jenkins:ls' locally
ls: Pulling from jenkins/jenkins
aa3e9ef32f73: Pull complete
a4b1a1e69b39: Pull complete
188bd27c22d7: Pull complete
43045d27df89: Pull complete
27090cc614a3: Pull complete
5fdc0b64433e: Pull complete
53373311d4de: Pull complete
ff65e9e2ceac: Pull complete
10655bddf081: Pull complete
4e813725ac37: Pull complete
ce55a371d187: Pull complete
7e33c265139d: Pull complete
Digest: sha256:f4f65e6cd1405cd889b7f5ac33f9d5cdc2a099de6b87fe8a3933b9c5d53d1d02
Status: Downloaded newer image for jenkins/jenkins:ls
1b3f551022b25199b5abc9732f0b223f527b6fe953421e57ea26eb0298f25ed1
```

Jenkins Initial Admin Login:

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

```
/var/jenkins_home/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

Installing Recommended Plugins:

Getting Started

Getting Started

✓ Folders	✓ OWASP Markup Formatter	✓ Build Timeout	✓ Credentials Binding	** commons-lang3 v3.x Jenkins API ** Ionicons API
🔌 Timestampmer	🔌 Workspace Cleanup	🔌 Ant	🔌 Gradle	Folders
🔌 Pipeline	🔌 GitHub Branch Source	🔌 Pipeline: GitHub Groovy Libraries	🔌 Pipeline Graph View	OWASP Markup Formatter
🔌 Git	🔌 SSH Build Agents	🔌 Matrix Authorization Strategy	🔌 LDAP	** ASM API
🔌 Email Extension	🔌 Mailer	🔌 Dark Theme		** JSON Path API
				** Struts
				** Pipeline: Step API
				** commons-text API
				** Token Macro
				Build Timeout
				** bouncycastle API
				** Credentials
				** Plain Credentials
				** Variant
				SSH Credentials
				Credentials Binding
				** SCM API
				** Pipeline: API

** - required dependency

Jenkins 2.568.1

Freestyle Project Creation:

Jenkins / All / New Item

New Item

Enter an item name

Flask-Freestyle

Select an item type

Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.

Organization Folder

OK

Successful Build Output: Triggered a build. The console output confirms successful cloning and workspace listing.

 Jenkins

Flask-Freestyle

#2

Status

Changes

Console Output

Delete build '#2'

Git Build Data

Edit build information

Previous Build

Console

Download

Copy

View as plain text

```
Started by user admin
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/Flask-Freestyle
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/Flask-Freestyle/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/aryan20s/Devops-Lab-L1-Submission.git # timeout=10
Fetching upstream changes from https://github.com/aryan20s/Devops-Lab-L1-Submission.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --progress -- https://github.com/aryan20s/Devops-Lab-L1-Submission.git +refs/heads/*:refs/remotes/origin/* #
timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 9c64aac236adfa42e476ee6bc00adf1bcc115932 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 9c64aac236adfa42e476ee6bc00adf1bcc115932 # timeout=10
Commit message: "Merge branch 'feature/user-auth' into main: Resolved conflict"
First time build. Skipping changelog.
[Flask-Freestyle] $ /bin/sh -xe /tmp/jenkins13499735263918853120.sh
+ ls -la
total 16
drwxr-xr-x 4 jenkins jenkins 4096 Jul 29 15:42 .
drwxr-xr-x 3 jenkins jenkins 4096 Jul 29 15:41 ..
drwxr-xr-x 8 jenkins jenkins 4096 Jul 29 15:42 .git
drwxr-xr-x 2 jenkins jenkins 4096 Jul 29 15:42 app
Finished: SUCCESS
```